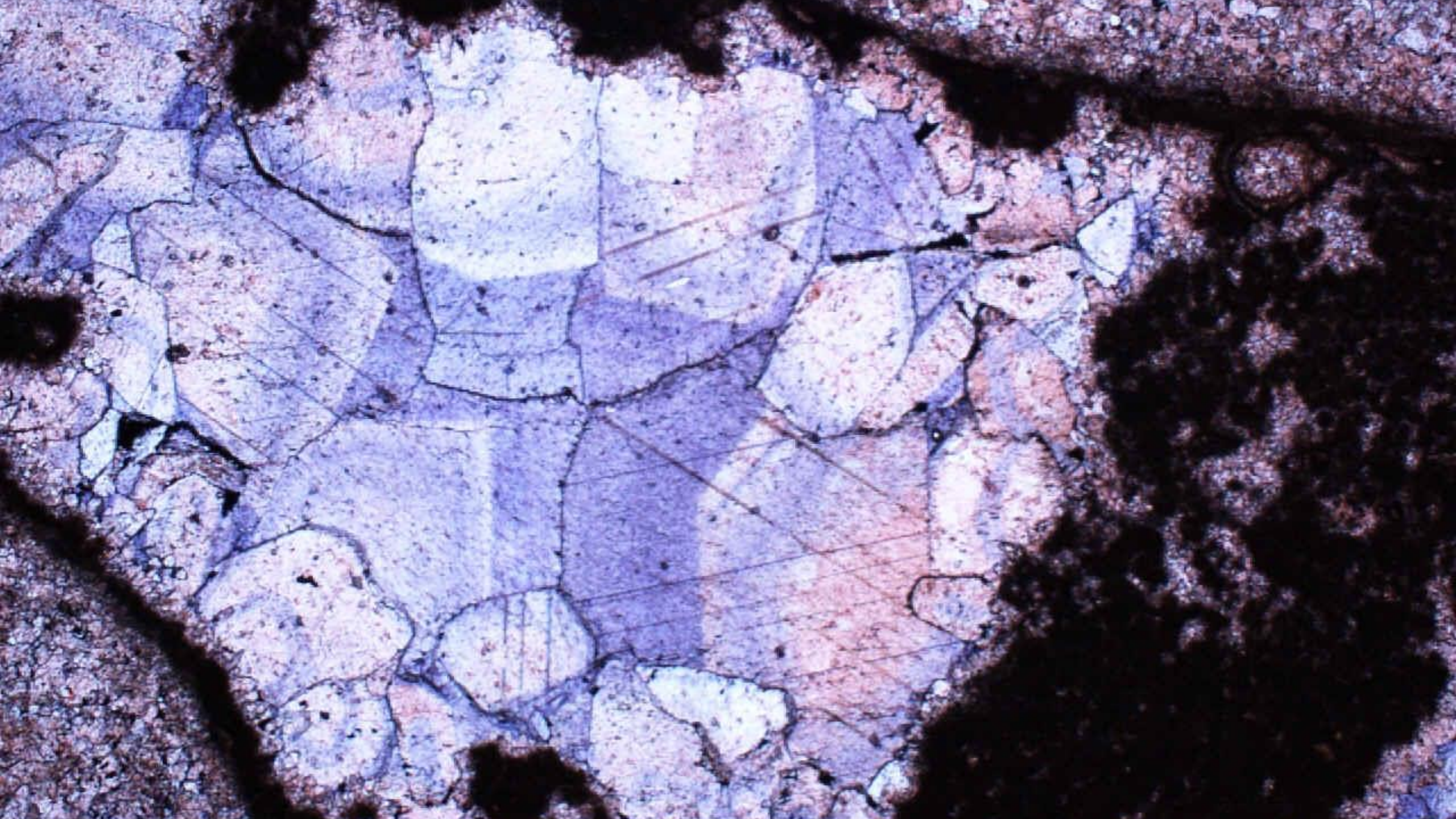
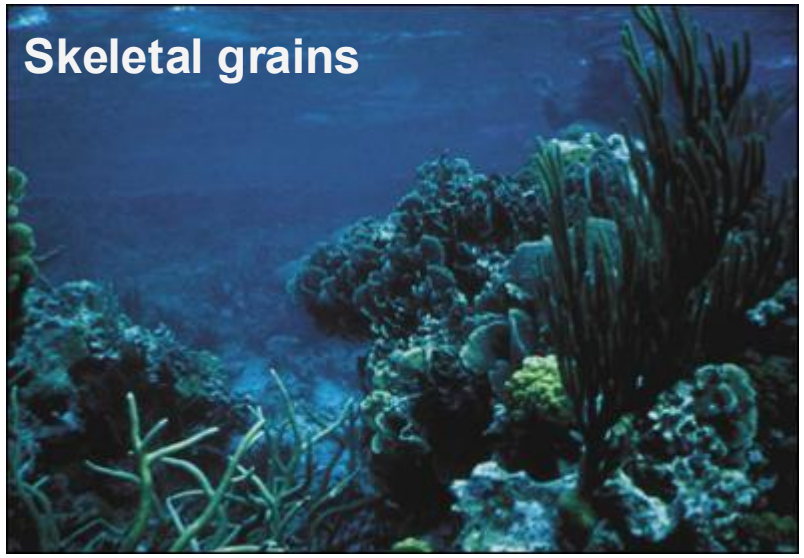
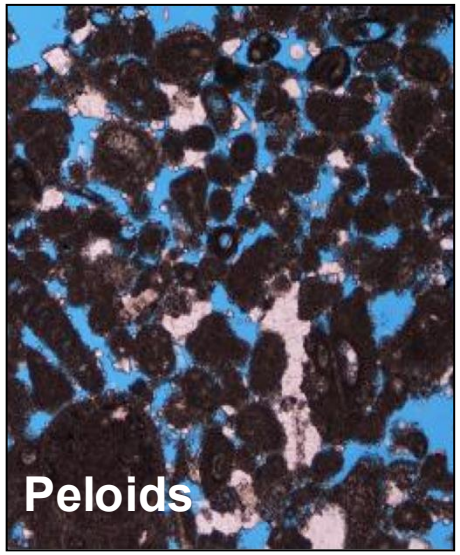


Principal components of carbonate rocks

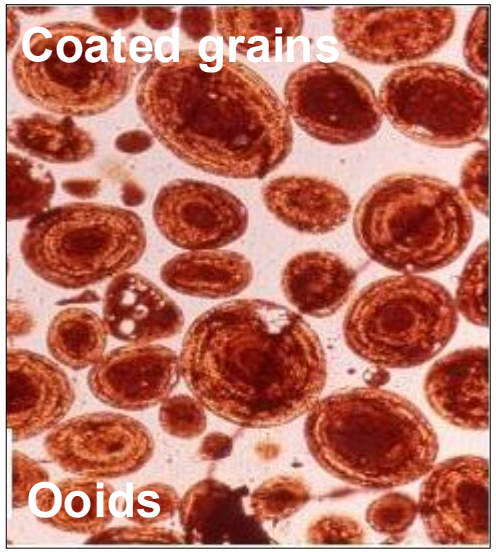


Courtesy Gregor Eberli

Non-skeletal grains











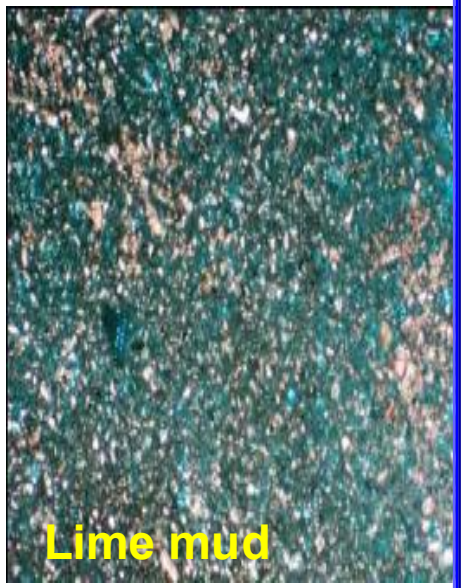


*Photo: C. Kendall/SEPM

Dunham classification (1/2)

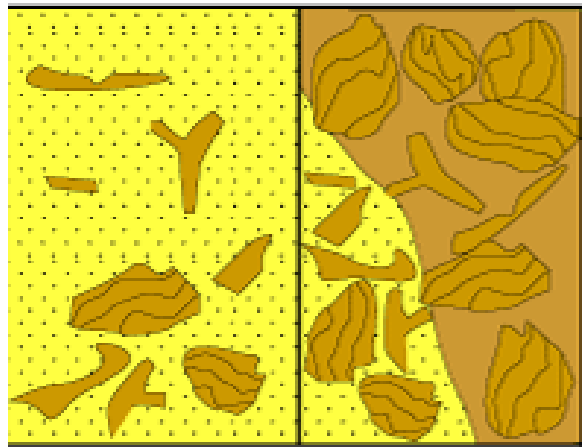
< 2mm allochems

Original components not bound together at deposition			
Contains mud (particles of clay and fine silt size)		Lacks Mud	
Mud-supported		Grain-supported	
Less than 10% Grains	More than 10% Grains		
Mudstone	Wackestone	Packstone	Grainstone



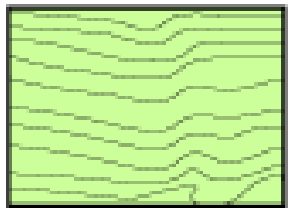
> 2mm allochems

Original components not bound organically at deposition	
>10% grains>2mm	
Matrix supported	Supported by >2mm component
Floatstone	Rudstone



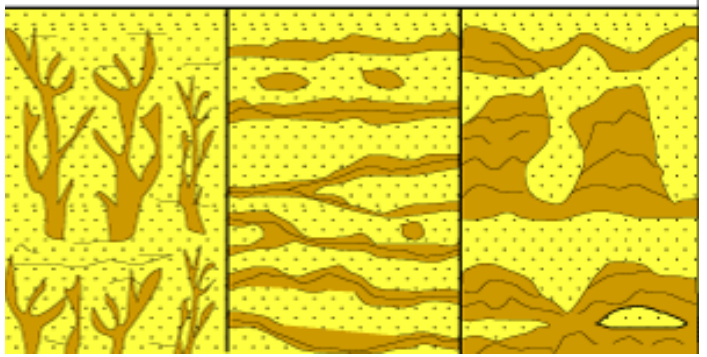
Dunham classification (2/2)

Boundstone





Autochthonous		
Original components bound organically at deposition		
By organisms that act as baffles	By organisms that encrust and bind	By organisms that build a rigid framework
Bafflestone	Bindstone	Framestone



- Based on depositional texture
- Better suited for rock descriptions using hand lens or binocular microscope
- Gives an idea of the porosity
- The most used in the petroleum industry

Exercise 1

We will use today to work through some carbonate thin sections, both as a review of carbonate petrography, as well as a bridge to the specific facies exercises we'll do rom next week. Keep in mind that for carbonate interpretation, you'll ideally combine hand samples (HS) and thin section (TS), and you need to familiarize yourself with how grain types look in HS and TS.

Today we'll study these fossil- and grain-rich samples:
L272
L337
L340

Take note of:
- Allochems, their %, size and degree of preservation
- Presence of sparite, micrite and/or porosity, and their %
- Texture
- What is the Dunham name for this thin section?
- Make sketches of key observations